

Proposed Industry Collaboration with South Cape College

To: South Cape College

Live proposal site: <https://superimmersive.github.io/south-cape-college-proposal/>

The live proposal site includes interactive examples, demonstration media and more detailed proposed Statement of Work pages for each training area. This document is intended as a formal printable overview of the same proposal.

Dear South Cape College Team,

I am writing on behalf of **Superimmersive** to propose the establishment of a working relationship with South Cape College, building on the positive engagement that has already taken place with the College's 4IR team.

I recently had the opportunity to present some of my existing immersive technology and VR work to members of the College's **4IR teaching and facilitation team**. The response to the technology was extremely positive, and I was subsequently invited to return the following day to demonstrate the work as part of the College's technology exhibition.

The exhibition provided an opportunity to demonstrate the technology to a wider audience, including representatives connected to the **Western Cape Premier's office**. During the exhibition, members of the visiting delegation were also able to experience the technology firsthand using a Meta Quest 3 headset and interact directly with one of my demonstration applications.

This initial engagement has demonstrated that there is already interest in the type of technology Superimmersive is developing and has provided a strong foundation from which to explore a more formal industry collaboration.

This proposal is therefore not a theoretical introduction to immersive technology, but a proposed next step following an initial demonstration and positive engagement with South Cape College's 4IR team.

I would like to propose beginning this relationship with a small **Proof of Concept (POC)** project based on a training requirement selected by South Cape College.

The purpose of the POC would be to demonstrate how immersive technology can be applied specifically to the College's existing curriculum and training environment, while allowing both parties to evaluate the potential for a longer-term working relationship.

The intention is not to replace existing curriculum or practical training, but to develop **custom immersive training experiences that complement the College's existing programmes** and provide learners with additional opportunities to interact with equipment, environments and processes in a controlled virtual environment.

Proposed Approach

We propose beginning with a **3-week Proof of Concept**, based on a training area selected by South Cape College.

The POC would allow both organisations to evaluate the technology, workflow and educational value of the collaboration before progressing into larger development scopes.

The objective is to establish a long-term relationship in which Superimmersive can serve as an **industry technology partner to South Cape College**, supporting the development of immersive training tools, demonstrations and other 4IR initiatives over time.

Benefits to South Cape College

1. Custom Training Solutions

Superimmersive can develop immersive applications specifically around South Cape College's training requirements.

These solutions would be designed as an **addition to the existing curriculum**, allowing learners to interact with equipment, environments and processes virtually before progressing to practical training in real-world facilities.

This can provide learners with additional opportunities to practise processes, understand systems and build familiarity before working with physical equipment.

2. Industry Exposure for Students

The collaboration can create opportunities for Gaming and IT students to gain practical experience through exposure to **real commercial projects and development workflows**.

Students could potentially participate in appropriate projects through practical placements, job shadowing and supervised industry exposure.

This would give learners an opportunity to see how skills developed within the College translate into commercial applications and industry workflows.

3. International Industry Exposure

The founder and director of Superimmersive has **11+ years of professional industry experience** across technical art, 3D development, software and immersive technology, with experience delivering work for major organisations and international clients.

This provides an opportunity to expose learners to **international industry standards, workflows and project requirements**, helping bridge the gap between academic training and the expectations of the global technology sector.

4. Strengthening South Africa's 4IR Private Sector

A partnership between South Cape College and a South African 4IR SME creates a direct connection between education and the private technology sector.

Supporting Superimmersive's development contributes to the growth of locally developed immersive technology, intellectual property, technical skills and employment opportunities.

The long-term objective is to develop immersive technology solutions for South Africa that can be deployed locally while also creating opportunities for **international work and export revenue**.

5. Marketing and Institutional Differentiation

Immersive technology provides South Cape College with highly visual and engaging demonstrations that can be used during campus visits, exhibitions, industry engagements and investor presentations.

Live VR and immersive training demonstrations can help showcase the College's 4IR capabilities and demonstrate how emerging technologies are being incorporated into education and skills development.

This can contribute to the College's positioning as an institution actively embracing emerging technology and industry collaboration.

6. Multiplatform Learning

Superimmersive can develop training experiences that are designed to be delivered across multiple platforms, allowing South Cape College to select the most appropriate technology for each training requirement.

Depending on the application, training content can be delivered through:

- **VR / Mixed Reality** — Fully immersive training experiences and mixed-reality simulations using VR headsets.
- **Web Browser** — Interactive web-based training experiences accessible through standard computers and mobile browsers without specialised hardware.
- **Mobile / AR** — Mobile applications and augmented-reality experiences accessible through smartphones and tablets.

This approach provides **greater accessibility and scalability**, allowing immersive training concepts to extend beyond learners who have access to VR hardware.

It also creates the potential for a single training concept to be adapted into multiple delivery formats depending on the requirements of the College, the available infrastructure and learner needs.

About Superimmersive

Superimmersive is a South African IT/4IR startup specialising in immersive technology, 3D development and interactive training experiences.

The company is **Black-owned and youth-owned**, with the founder born in 1992.

The founder and director has **11+ years of professional industry experience** across technical art, 3D development, software and immersive technology, with experience working with major organisations and industries in South Africa and internationally.

Selected organisations associated with previous professional work include:

Absa, Anglo American, Aston Martin, Blair Project / ProtoEV, Budweiser, Capel Manor College, Cadbury, Catapult Energy Systems, Coventry University, Darlington Libraries, Game South Africa, Hasbro, HP, HRUC, Investec, Johnnie Walker, Network Rail, Nedbank,

Nissan, Roche Diagnostic Solutions, SANRAL, Sasol, Thule, Transnet, Tyne Coast College, UNIDO, University of Johannesburg, Weylandts and Nestlé.

This work was delivered across studio engagements, contract roles and other professional contexts over an 11+ year career. It is not a claim that each organisation listed is a direct Superimmersive client.

This experience provides Superimmersive with a combination of **technical development capability, commercial experience and exposure to international industry requirements.**

Our Mission

Superimmersive's long-term mission is to become an industry leader in **immersive technology and immersive training tools**, providing South African educational institutions with access to high-quality, industry-standard immersive training experiences.

Our goal is to help equip learners with practical industry knowledge and familiarity with complex equipment, processes and environments **before they enter real-world facilities.**

We also aim to contribute to the growth of South Africa's technology sector by developing locally owned intellectual property, creating employment opportunities and connecting South African technical skills with international markets.

A key part of this vision is to develop **licensed immersive training tools in collaboration with universities and training institutions**, with each solution designed around the institution's specific curriculum and training requirements.

Proposed Proof of Concept

As the next step in the relationship, Superimmersive proposes developing a **3-week Proof of Concept application** based on a training requirement selected by South Cape College.

The College may select one of the suggested Proof of Concept options below, or propose an alternative training requirement for consideration.

Each area below leads with the **suggested Proof of Concept** currently detailed on the live proposal site. Later expansion is listed as a potential next step, not as part of the 3-week POC.

The final POC scope, functionality, development timeline and cost will be agreed before development begins.

Welding Training

Suggested POC (Meta Quest 3): A focused VR welding exercise. The learner picks up a virtual welding tool, positions the rod against a workpiece, receives basic feedback (sound, sparks and simulated weld deposits), observes rod wear, has the welding path recorded, and can view a **2D representation of that path** for later assessment.

Potential later expansion: Welding equipment and safety, workstation setup, guided exercises, multiple welding processes, performance scoring and learner progress tracking.

Assembly-Line & Pneumatic Systems

South Cape College already has **pneumatic scale models representing assembly-line operations**. The suggested POC uses that as context, without attempting a complete digital twin of the physical equipment.

Suggested POC (Meta Quest 3): A guided virtual assembly and sorting exercise. The learner selects components from a UI menu, assembles a simplified **separator system** using visual ghosted guides, places objects onto a conveyor, starts the line, observes sorting to matching destinations, and receives basic outcome feedback.

Potential later expansion: Unguided assembly assessment, pneumatic system interaction, PLC wiring and configuration, process sequencing, fault finding and a closer digital extension of the College's existing physical training equipment.

PPE Training

Suggested POC (Meta Quest 3): PPE application onto a digital mannequin. The learner selects items from a focused set, applies them to the mannequin, receives a simple correctly / incorrectly applied outcome, and can reset and repeat.

Potential later expansion: PPE identification and selection, application across multiple tasks, and compliance assessment.

Tool Assembly & Disassembly

Suggested POC (Meta Quest 3): Tool component identification and replacement. The learner expands an interactive 3D tool, inspects components with descriptions and voice-over, then completes a guided task: disassemble with supplied virtual tools, replace a specified component, and reassemble in the correct sequence.

Potential later expansion: Additional tools and components, multiple assembly and disassembly procedures, guided and unguided exercises, fault finding, maintenance procedures and learner assessment.

Electrical Training

Suggested POC (Meta Quest 3): A basic lighting circuit. The learner identifies the required components, connects terminals between a **distribution board, light switch and light fitting**, creates virtual wires, receives feedback on incorrect connections, completes the circuit, activates the light, and operates the switch.

Potential later expansion: Additional circuit types, unguided wiring, circuit-diagram interpretation, fault finding, electrical testing and electrical safety.

Solar Technology

Suggested POC (Meta Quest 3): Solar generation and panel configuration. The learner identifies key components, adjusts panel **orientation and angle**, controls simulated **sun position**, switches **Sunny / Cloudy** conditions, and compares generation output.

Potential later expansion: Multiple panels and array configuration, inverters, battery storage, electrical connections, system sizing, fault finding and maintenance.

Wind Turbine Technology

Suggested POC (Meta Quest 3): Wind-turbine operating conditions. The learner controls **wind speed and direction**, observes the effect on generation, and reads a simple power-output indicator.

Potential later expansion: Turbine components, broader operating-condition scenarios and performance analysis.

EV Technology

Suggested POC (Meta Quest 3): EV motor inspection and component replacement. The learner expands an interactive 3D motor, inspects components with descriptions and voice-over, then completes a guided task: disassemble with supplied virtual tools, replace a specified part, and reassemble in the correct sequence.

Potential later expansion: Additional maintenance procedures, battery systems, diagnostics, high-voltage safety and fault-finding exercises.

Hydrogen Technology

Suggested POC (Meta Quest 3): Electrolyser operation and hydrogen production. The learner inspects a simplified electrolyser, confirms water supply and electrical input, starts electrolysis, observes water separating into **hydrogen and oxygen**, adjusts a simplified input such as power, then shuts down and resets.

Potential later expansion: More detailed electrolyser operation, safety procedures, process controls, hydrogen storage and compression, fuel-cell systems and broader hydrogen-energy production workflows.

Tourism

Suggested POC (Web): A browser-based **3D Gaussian splat (3DGS)** explore. The learner explores a single captured environment, selects marked points of interest, opens an information panel and hears accompanying voice-over.

Potential later expansion: Additional digital site tours, 3D / 360° experiences and guided virtual tourism modules. Capture of a College-selected tourism site would be scoped separately if required.

Administration / Business Processes

Suggested POC (Web): One **AI voice-driven customer-service call**. The learner starts a simulated workplace conversation in the browser, speaks with an AI-driven simulated customer, follows one focused conversation path, then completes or resets and repeats.

Potential later expansion: Additional workplace interactions, customer-service scenarios and business-process simulations.

POC Scope, Timeline & Cost

The selected concept would first be refined into a clearly defined Proof of Concept scope, including the **core functionality, interactions and intended learning outcomes**.

A standard Proof of Concept is expected to take approximately **3 weeks to develop, at a fixed development cost of R85,000**.

The final POC scope will be agreed before development begins to ensure that the project remains focused and achievable within the agreed timeline and budget.

Any hardware or specialist equipment required for the selected POC will be assessed separately and **quoted independently from the software development**.

The POC is intended to demonstrate the core concept, interaction and educational potential of the proposed solution. It should not be considered a complete production-ready training system unless specifically agreed as part of the scope.

Payment Terms

Payment of the full **R85,000 Proof of Concept fee is required prior to commencement of development**.

Where South Cape College's procurement or payment processes require an alternative payment arrangement, this can be discussed and agreed upon prior to commencement of the project.

Long-Term Opportunity

The Proof of Concept is intended to be the **first step rather than the end goal**.

If successful, the relationship could progress into larger projects involving the development of multiple immersive training modules, curriculum-specific applications, student industry exposure and the development of commercially licensable training tools.

For Superimmersive, establishing South Cape College as a direct institutional client would represent an important milestone in transitioning from primarily third-party/client-based work toward direct partnerships with South African educational institutions.

This relationship would also strengthen Superimmersive's ability to pursue further investment and funding opportunities, including programmes such as the **Technology Innovation Support (TIS) programme**, enabling the business to invest in higher-end equipment, workspace, human resources and the capacity required to undertake larger projects.

This would ultimately allow Superimmersive to take on larger projects, improve turnaround times, create opportunities for students to gain industry exposure and expand South African immersive technology capabilities into international markets.

Proposed Partnership

Superimmersive would like to explore a relationship with South Cape College based on the following progression:

Proof of Concept → Evaluation → Larger Training Projects → Long-Term Industry Partnership

The initial technology demonstration and engagement with the College's 4IR team has already provided an opportunity to demonstrate the capabilities of Superimmersive and establish initial interest.

The proposed POC would therefore serve as the **next practical step**.

It provides South Cape College with a relatively low-risk opportunity to experience the technology in the context of one of its own training requirements and evaluate its potential educational value before progressing to larger development scopes.

Superimmersive brings **industry experience, technical expertise and international commercial exposure**, while South Cape College brings its **educational expertise, curriculum knowledge, training facilities and learner base**.

The combination of these capabilities has the potential to create meaningful opportunities for learners while contributing to the development of South Africa's emerging 4IR technology sector.

I look forward to building on our initial engagement and working with South Cape College to identify and develop a suitable first Proof of Concept.

Kind regards,

Devon Kirchner
Founder & Director
Superimmersive

Email: devon@superimmersive.io

Phone: 068 594 3568

Web: <https://superimmersive.github.io/>

Location: George, Western Cape